Topology Optimization of Planar Elastic Energy-Storing Structures in a Fixed Area

Lab III – Project Proposal

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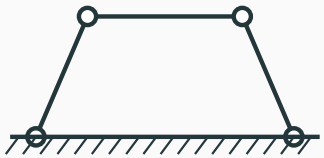
Compliant vs. Traditional Mechanisms

Traditional (rigid-body)

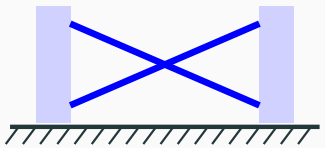
- Rigid links + discrete joints (pin, slider, bearing)
- Motion from assembled kinematic pairs
- Friction, backlash, wear; needs lubrication
- Multi-part assembly; energy stored externally (coil spring, torsion bar)

Compliant (flexure-based)

- Single monolithic part; motion = elastic deformation [1]
- No sliding contacts ⇒ no friction or wear
- Backlash-free, miniaturisable to MEMS [2]
- Energy stored *in* the structure itself



Pin-jointed linkage

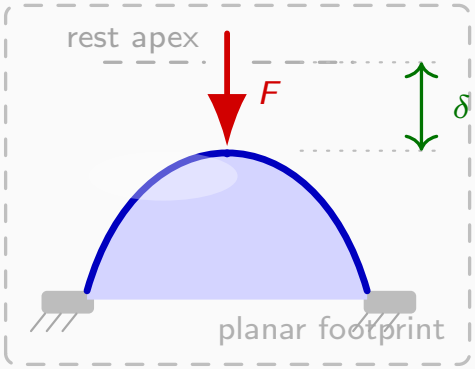


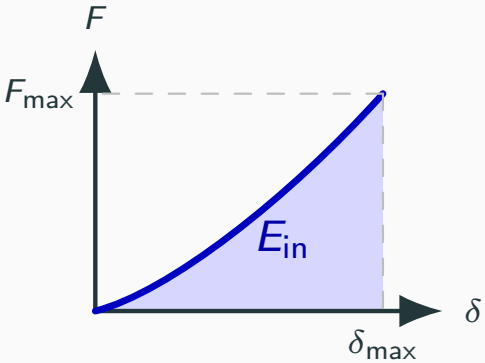
Compliant flexure pivot

Background: Planar Elastic Energy Storage

- Compliant flexures store and release energy via reversible elastic deformation [1].
- Applications: MEMS actuators, soft grippers, watch springs, miniature energy harvesters [2].
- Often confined to thin planar footprints (PCB-mounted, wearable, additively manufactured) [3].
- Key metric — **energy storage efficiency**:

$$\eta = \frac{E_{\text{out}}}{E_{\text{in}}} = \frac{\int F_{\text{out}}\ \mathrm{d}x}{\int F_{\text{in}}\ \mathrm{d}x}$$





Motivation & Research Gap

- Within a fixed planar footprint, geometry dictates the load path and strain distribution, hence the stored elastic energy at a given material volume [1].
- Folded / serpentine layouts lengthen the effective beam inside the same area, distributing strain more uniformly and delaying yield at the maximum-stress fibre.
- Topology optimisation (SIMP) generates non-intuitive material layouts that minimise compliance under a volume constraint [4], [5], [6].
- **Gap:** experimental comparisons of fold count and pattern family at *equal volume* and *equal footprint* are scarce; most topology-optimised planar springs are reported only numerically [2], [3].

Project Goals

1. **Define objective function** for elastic energy storage performance within a fixed planar footprint ($100 \times 100 \times 3\,\mathrm{mm}$).
2. **Generate candidate structures** via:
   - Manual baseline designs (straight, folded, serpentine, curved beams)
   - Computational topology optimisation (SIMP method)
3. **Compare** force–displacement behaviour, energy storage E_{in}, energy release E_{out}, and efficiency η across topologies.
4. **Analyse** how topology affects load paths, deformation modes, and stiffness distribution.

Hypotheses

- **H1 — Fold-count effect within a pattern family:** For a fixed planar footprint ($100 \times 100 \times 3\,\mathrm{mm}$) and constant material volume, the elastic energy storage efficiency η of a folded beam varies systematically with the *number of folds* within a given pattern (e.g. a triangular wave with 1, 2, 3, ... folds; a square wave across multiple folds).
- **H2 — Topology optimisation:** Topology-optimised structures will outperform manually designed baselines in terms of energy efficiency and performance stability.

Experimental Variables

	H1: Fold count / pattern	**H2: Topology optimisation**
Independent	Number of folds n Pattern family (triangular wave, square wave, ...)	Design method (manual baseline vs. SIMP-optimised)
Dependent	η, E_{in}, E_{out}, stiffness k, peak force F_{max}	η, E_{in}, E_{out}, stiffness k, trial-to-trial variance
Controlled	Material volume Footprint $100\times100\times3$ mm PLA + print parameters Clamping BCs Quasi-static loading rate	Volume fraction f Footprint $100\times100\times3$ mm PLA + print parameters Clamping BCs Quasi-static loading rate

Theory Framework

The theoretical basis is divided into three parts:

1. **Topology optimization model**: convert structure design into a material distribution problem.
2. **Mechanical constraints**: ensure the design is safe under static and cyclic loading.
3. **Energy-based objective**: optimise stored elastic energy and energy release efficiency.

Topology Optimization Model

- The design domain is divided into finite elements.
- Each element has a density variable:

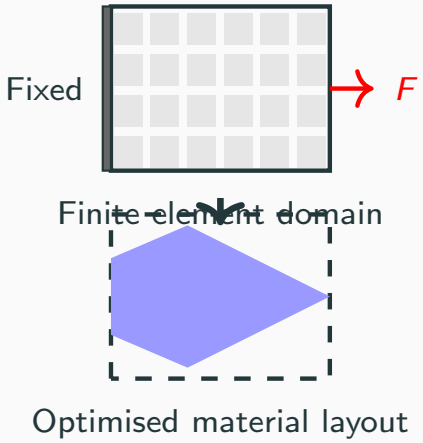
$$\rho_e \in [0, 1]$$

- $\rho_e = 1$ represents solid material, while $\rho_e = 0$ represents void.
- The SIMP method connects density and material stiffness [4], [5]:

$$E_e = E_{\min} + \rho_e^p (E_0 - E_{\min})$$

- The structural response is obtained from:

$$\mathbf{K}(\boldsymbol{\rho})\mathbf{U} = \mathbf{F}$$



Mechanical and Physical Constraints

- The optimised structure must be mechanically feasible, not only numerically optimal.
- **Static strength**: avoid yielding or fracture:

$$\sigma_{\mathrm{vm}} \leq \sigma_y$$

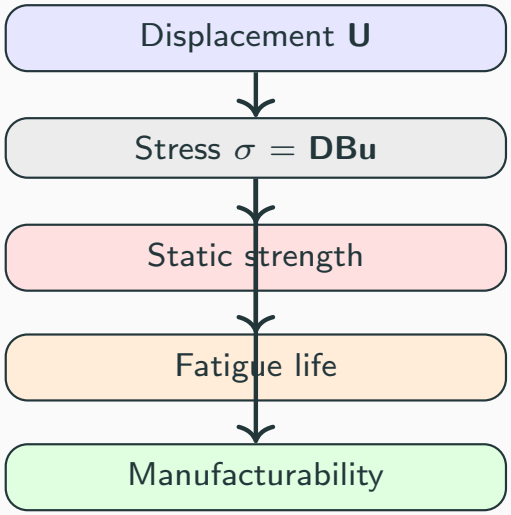
- **Fatigue constraint**: evaluate cyclic loading using a Goodman-type criterion **zhao2023topology**:

$$\frac{\sigma_a}{\sigma_e} + \frac{\sigma_m}{\sigma_{ut}} \leq 1$$

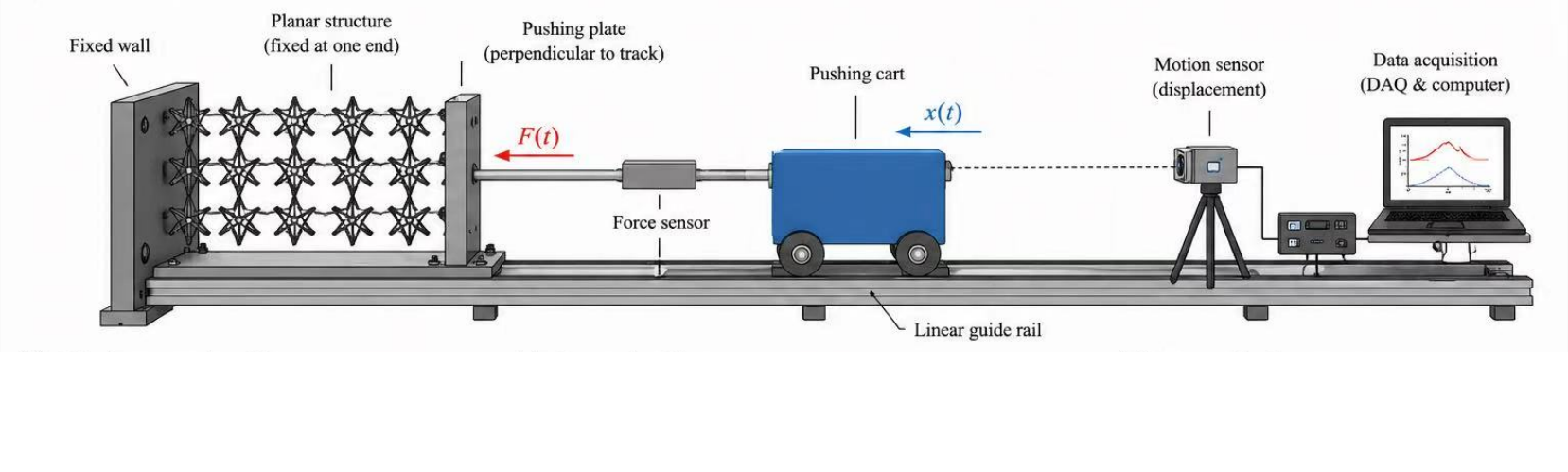
- Mean and alternating stresses are:

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}, \qquad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$$

- **Manufacturing**: density filtering



Experimental Setup



Specimen: planar spring structure clamped at one end

Force sensor: records applied load $F(t)$ during compression

Motion sensor + DAQ: records displacement $x(t)$ and synchronises acquisition

Measurement Workflow

Quasi-static manual compression of specimen → Record synchronized $F(t)$ and $x(t)$ from the sensors → Convert to F–x and integrate area for E_{in}, E_{out}, η

- **Loading method:** hand-wheel actuation with steady manual input.
- **Quality check:** near-linear $x(t)$ confirms a stable quasi-static loading rate.
- **Output:** force–displacement curves used to compare energy storage and release behaviour.

Apparatus & Test Conditions

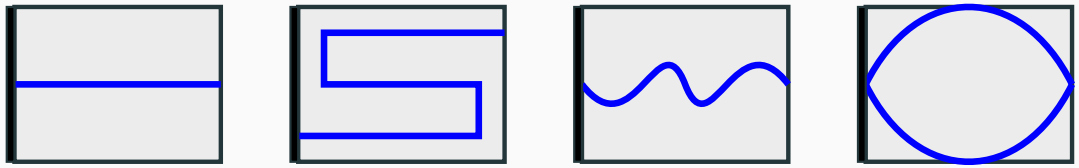
Rig & sensing

- Custom hand wheel + lead screw compression rig
- Linear guide rail and pushing plate
- Force sensor for applied load $F(t)$
- Motion sensor for displacement $x(t)$
- One-end clamped boundary condition

Specimens & analysis

- FDM-printed PETG planar test structures
- Fixed footprint and equal-thickness specimens
- GNU Octave, FreeCAD, and Python for design / analysis
- Room temperature ($\sim 25\,^{\circ}\mathrm{C}$)
- Quasi-static manual loading for force–displacement evaluation

Candidate Structures



(a) Straight (b) Folded (c) Serpentine (d) Curved

Four manual baselines + 1–2 topology-optimised candidates. All share identical footprint, thickness, and boundary conditions.

- **Design space:** 4 manual baselines vs. 1–2 topology-optimized candidates.
- **Constraints:** Identical footprint ($W \times L$), thickness, and PETG material.
- **Goal:** To evaluate the influence of geometric topology on energy storage E.

Analysis Plan

1. **Data acquisition:** Record force F and displacement δ at 0.1 mm increments during quasi-static compression. Repeat each topology ≥ 3 times.
2. **Preprocessing:** Zero-offset correction; apply force sensor calibration curve; discard first cycle (pre-conditioning).
3. **Feature extraction:**
   - Stiffness: slope of linear region of F–δ curve.
   - E_{in}: area under loading curve (trapezoidal integration).
   - E_{out}: area under unloading curve.
   - $\eta = E_{\text{out}}/E_{\text{in}}$.
4. **Statistical treatment:** Mean $\pm$ standard deviation across trials; compare topologies using bar charts with error bars.
5. **Validation:** Compare experimental stiffness with FEA prediction (if time permits, COMSOL cross-check).

Safety & Compliance

Main hazards

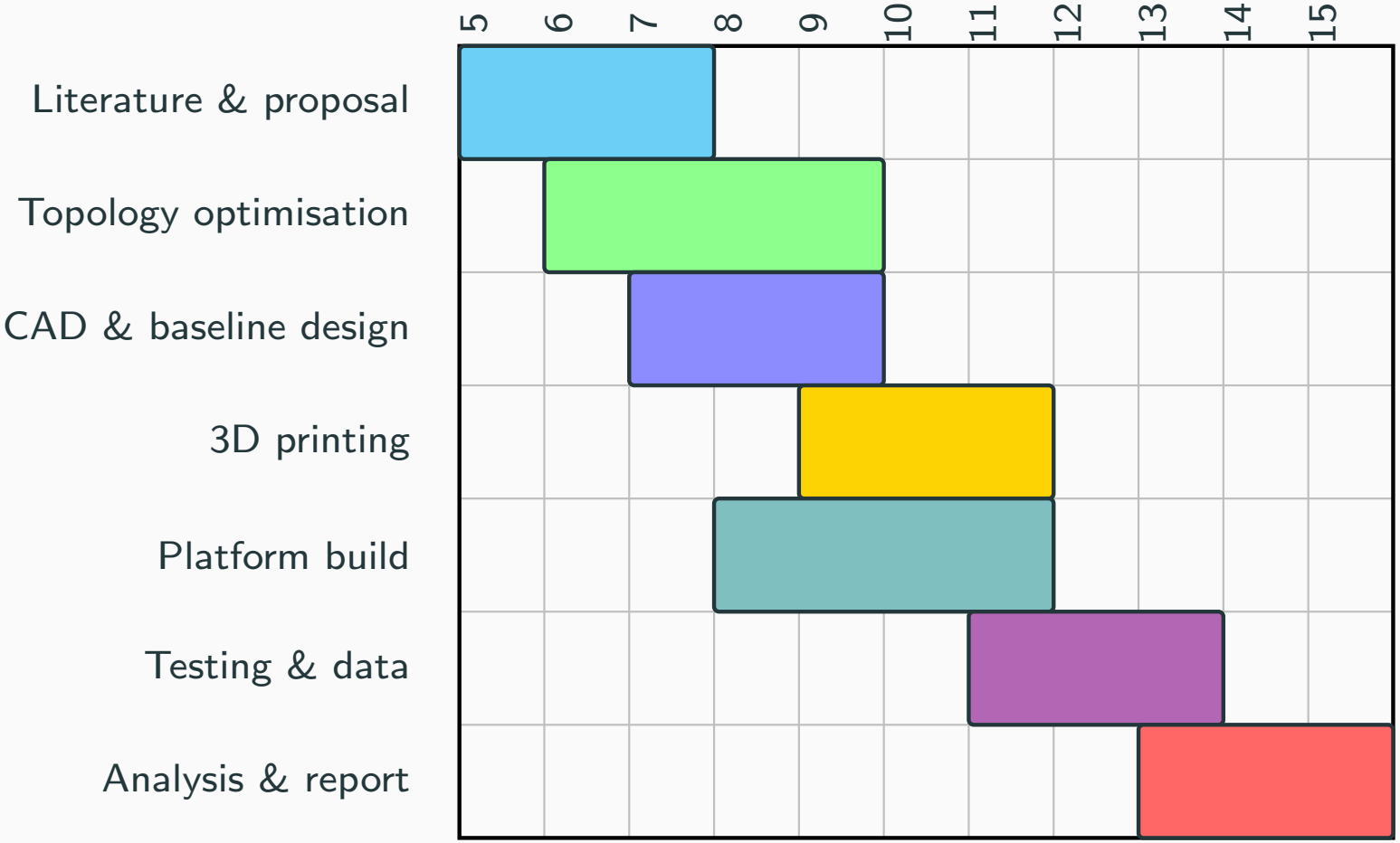
- Hot nozzle / bed and moving printer axes
- Sharp edges after support removal or trimming
- Pinch points at the lead screw and moving carriage
- Sudden spring-back or cracking of polymer samples
- Sensor cables snagging on the rig

Controls

- Ventilated printing; allow printer to cool before handling
- Deburr samples; keep hands clear of compression path
- Wear safety glasses during every loading test
- One operator applies load slowly and stops at slip, crack, or permanent set
- Secure specimen / sensors and route cables clear

Compliance: Benchtop mechanical experiment only; no chemical, radiation, or biological hazards. **Waste:** Failed prints and broken polymer samples are disposed as solid plastic waste. **Emergency:** Lab supervisor, makerspace / lab technician, campus security (ext. 8888).

Project Timeline



Budget

Item	Qty	Unit (MYR)	Total (MYR)
Materials & Consumables			
Filament for specimens (1 kg spool)	1	40	40
Spare filament & reprints	–	–	20
Sandpaper / tape / small tools	–	–	10
Total			**70**

References i

[1] L. L. Howell, *Compliant Mechanisms*. New York: John Wiley & Sons, 2001.

[2] B. Zhu et al., "Design of compliant mechanisms using continuum topology optimization: A review," *Mechanism and Machine Theory*, vol. 143, p. 103 622, 2020. DOI: 10.1016/j.mechmachtheory.2019.103622

[3] X. Tong, Q. Ge, et al., "Design of planar springs for compliant energy storage mechanisms via topology optimization," *Journal of Mechanical Design*, vol. 145, no. 5, p. 051 701, 2023. DOI: 10.1115/1.4056959

[4] M. P. Bendsøe and O. Sigmund, *Topology Optimization: Theory, Methods, and Applications*. Berlin, Heidelberg: Springer, 2003. DOI: 10.1007/978-3-662-05086-6

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[5] O. Sigmund, "A 99 line topology optimization code written in Matlab," *Structural and Multidisciplinary Optimization*, vol. 21, no. 2, pp. 120–127, 2001. DOI: 10.1007/s001580050176

[6] E. Andreassen, A. Clausen, M. Schevenels, B. S. Lazarov, and O. Sigmund, "Efficient topology optimization in MATLAB using 88 lines of code," *Structural and Multidisciplinary Optimization*, vol. 43, no. 1, pp. 1–16, 2011. DOI: 10.1007/s00158-010-0594-7